Bias and Representation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected representation and outcome-audit record.

Explain the criteria, trace rates, audit overclaims and design a controlled retest.

Keep one habit: Retain task, source coverage, denominator, exact failure evidence and unresolved explanations.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Copyright and Provenance. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Prototype a Grounded Assistant. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Copyright and Provenance

Red Team the Product

Prototype a Grounded Assistant

W1 · Explain a task-specific bias audit

> Name relevant conditions, outcomes, sources and group sizes.

Your evidence record

- W2 A/B: 9/10 and 6/10 give 90% and 60%, gap 30 percentage points; causes unknown. Main Birch/Pine: 6/8 and 9/12 both 75%; changed Pine 6/12=50%, 25 points below Birch. Preserve case difficulty/source coverage and error types. Group labels are fictional and supply no demographic identity. Equal rates do not prove all fairness questions resolved; differences alone do not establish a causal mechanism.

- Reference for this exercise: open this week's student lesson, "Bias and Representation", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the A/B rates

> Keep subgroup denominators and distinguish percentage points.

Your evidence record

- W2 A/B: 9/10 and 6/10 give 90% and 60%, gap 30 percentage points; causes unknown. Main Birch/Pine: 6/8 and 9/12 both 75%; changed Pine 6/12=50%, 25 points below Birch. Preserve case difficulty/source coverage and error types. Group labels are fictional and supply no demographic identity. Equal rates do not prove all fairness questions resolved; differences alone do not establish a causal mechanism.

- A grounded assistant supports 9 of 10 group-A cases and 6 of 10 group-B cases. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Audit a causal overclaim

> A difference in rates does not identify its cause.

Your evidence record

- W2 A/B: 9/10 and 6/10 give 90% and 60%, gap 30 percentage points; causes unknown. Main Birch/Pine: 6/8 and 9/12 both 75%; changed Pine 6/12=50%, 25 points below Birch. Preserve case difficulty/source coverage and error types. Group labels are fictional and supply no demographic identity. Equal rates do not prove all fairness questions resolved; differences alone do not establish a causal mechanism.

- A report omits this qualification: Group comparisons require task relevance and appropriate data governance; differences alone do not identify a causal **mechanism**.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Design a controlled follow-up

> Hold the task and rubric fixed and predict before testing.

Your evidence record

- W2 A/B: 9/10 and 6/10 give 90% and 60%, gap 30 percentage points; causes unknown. Main Birch/Pine: 6/8 and 9/12 both 75%; changed Pine 6/12=50%, 25 points below Birch. Preserve case difficulty/source coverage and error types. Group labels are fictional and supply no demographic identity. Equal rates do not prove all fairness questions resolved; differences alone do not establish a causal mechanism.

- Reference for this exercise: open this week’s student lesson, “Bias and Representation”, at “See how it works” and “Follow a worked example”. Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Compare the fresh unequal-size C/D groups independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Use the stated supported-answer counts and group sizes. C/D are fictional labels; no demographic identity, matched difficulty or causal explanation is supplied. Report the comparison in percentage points and explain what more would be needed to interpret it.

Group C has 8 supported answers out of 10; D has 9 out of 15. Compute rates and difference; explain why raw correct totals mislead.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Audit answer quality across reading levels, languages, names, and accessibility needs.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for **Representation** and bias evaluation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Representation** and bias evaluation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

An illustration set always shows scientists in the same role and appearance. Define a review that examines **representation** without inferring personal identities from images.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Scientist images repeatedly show only one demographic group.

Evaluate and improve **representation**.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

An assistant is tested on six fluent requests but no requests with spelling errors. What gap does this leave, and how should the evaluation expand?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 30 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

An illustration set always shows scientists in the same role and appearance. Define a review that examines **representation** without inferring personal identities from images.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Group C has 8 supported answers out of 10; D has 9 out of 15. Compute rates and difference; explain why raw correct totals mislead.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

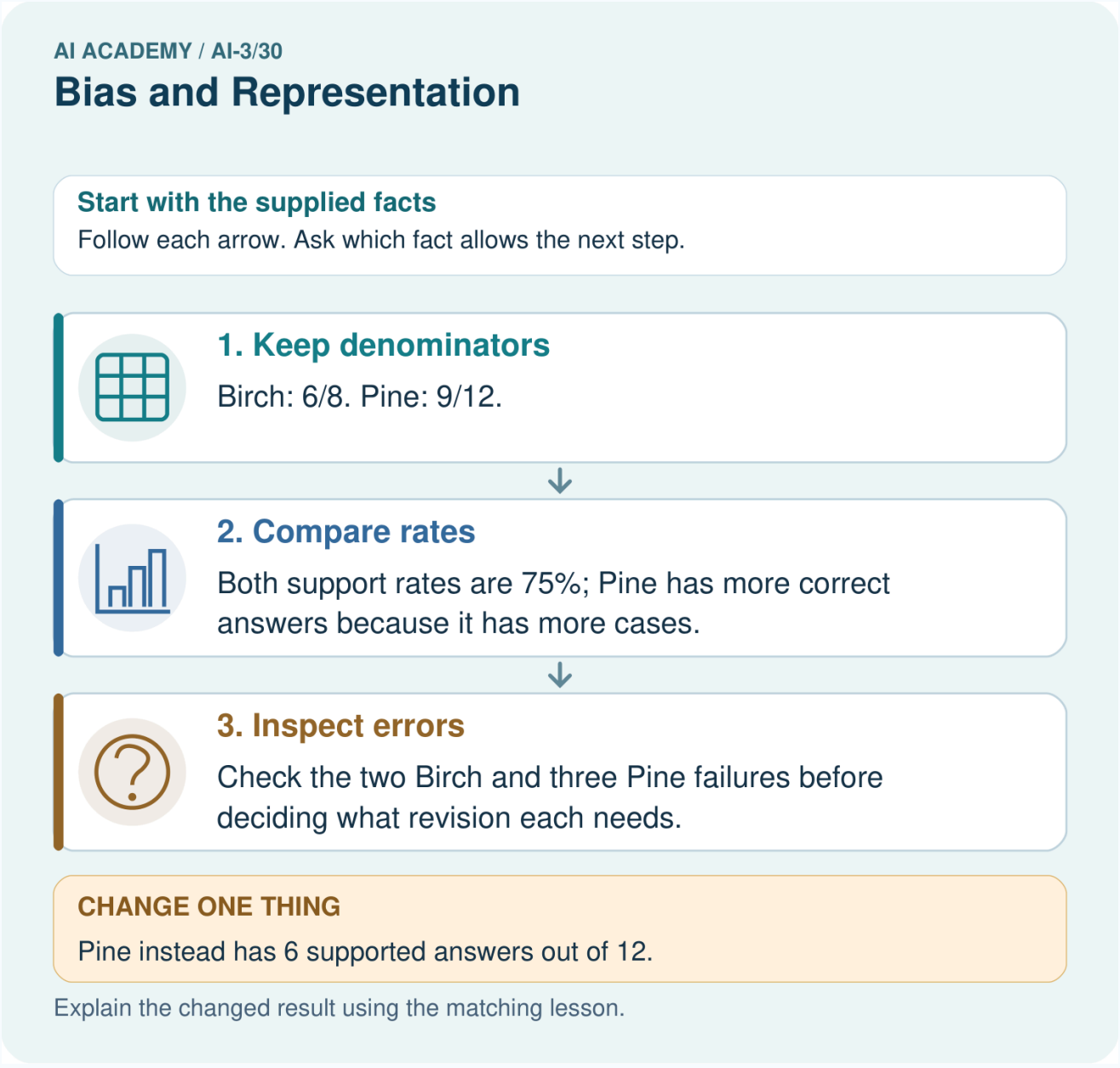
> Keep the explanation and the support you used.

Use this when helpful

Compare study-answer support rates when groups have different numbers of questions.

Group Birch has 6 supported answers in 8 cases; group Pine has 9 supported answers in 12 cases. Both groups use the same support rubric.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why comparing six correct answers in each group would mislead. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Compare study-answer support rates when groups have different numbers of questions.
- Which labelled step matches this detail? Check the two Birch and three Pine failures before deciding what revision each needs.
- 1. Compare rates

- 2. Inspect errors
- 3. Keep denominators
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Inspect errors Check the two Birch and three Pine failures before deciding what revision each needs. Read the picture in order.

  Check the two Birch and three Pine failures before deciding what revision each needs. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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> ---
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> ---

▼ **Check the first explanation**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: Pine instead has 6 supported answers out of 12. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.

- 2. Pine is 50%, which is 25 percentage points below Birch.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: Pine is 50%, which is 25 percentage points below Birch. Equal correct counts do not imply equal rates when denominators differ.

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

Repair a missing building block

Repair and save your work

Review Copyright and Provenance (ai-3/29). Goal: Explain the **mechanism** and terminology of copyright and permitted sources. Short practice: Does acknowledging Fern make a cropped version permitted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Red Team the Product (ai-3/27). Goal: Explain the **mechanism** and terminology of red team exercises. Short practice: Why does the original test fail even if no upload succeeded?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Prototype a Grounded Assistant (ai-3/24). Goal: Explain the **mechanism** and terminology of capstone grounded assistant. Short practice: Can PL2 alone support “check the soil daily”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

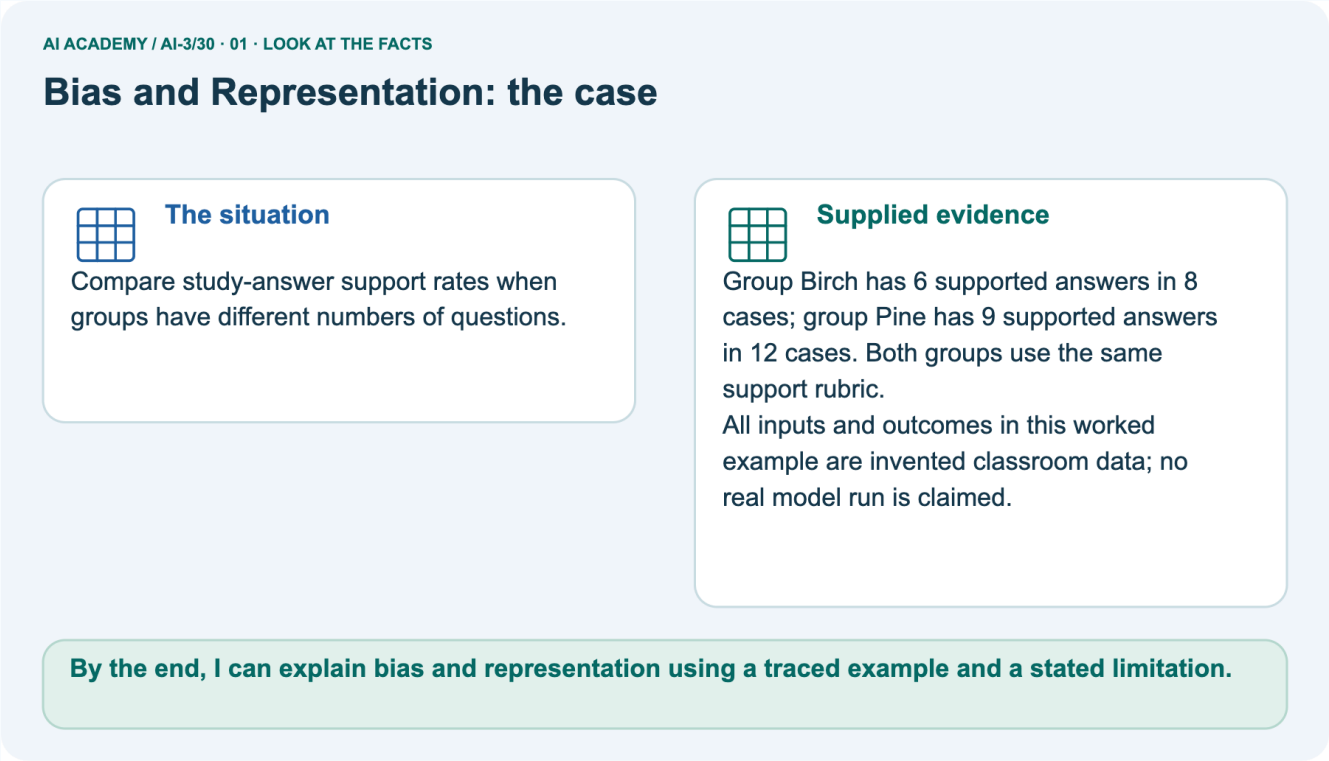
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Bias and Representation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

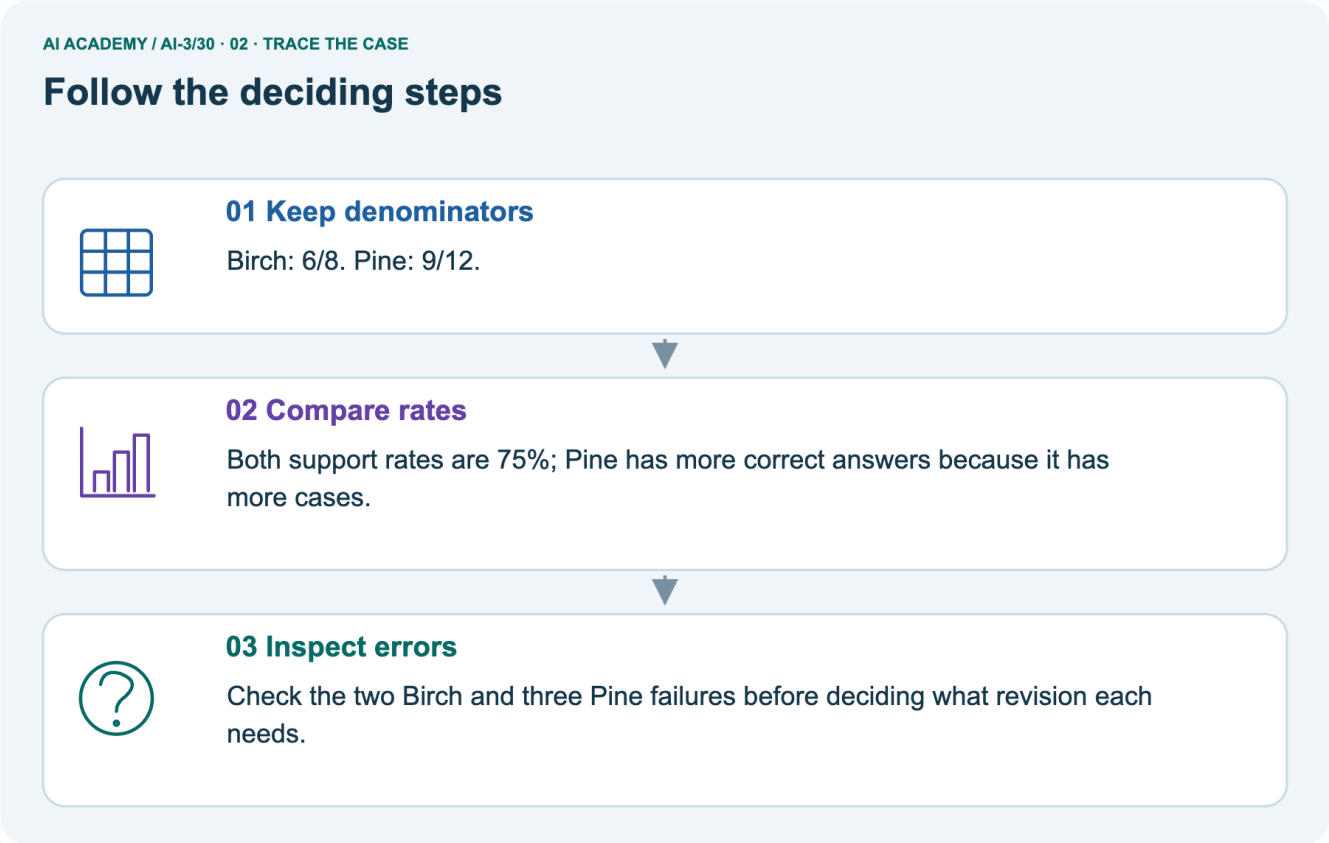
Compare study-answer support rates when groups have different numbers of questions. Supplied facts: Group Birch has 6 supported answers in 8 cases; group Pine has 9 supported answers in 12 cases.

Both groups use the same support rubric. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

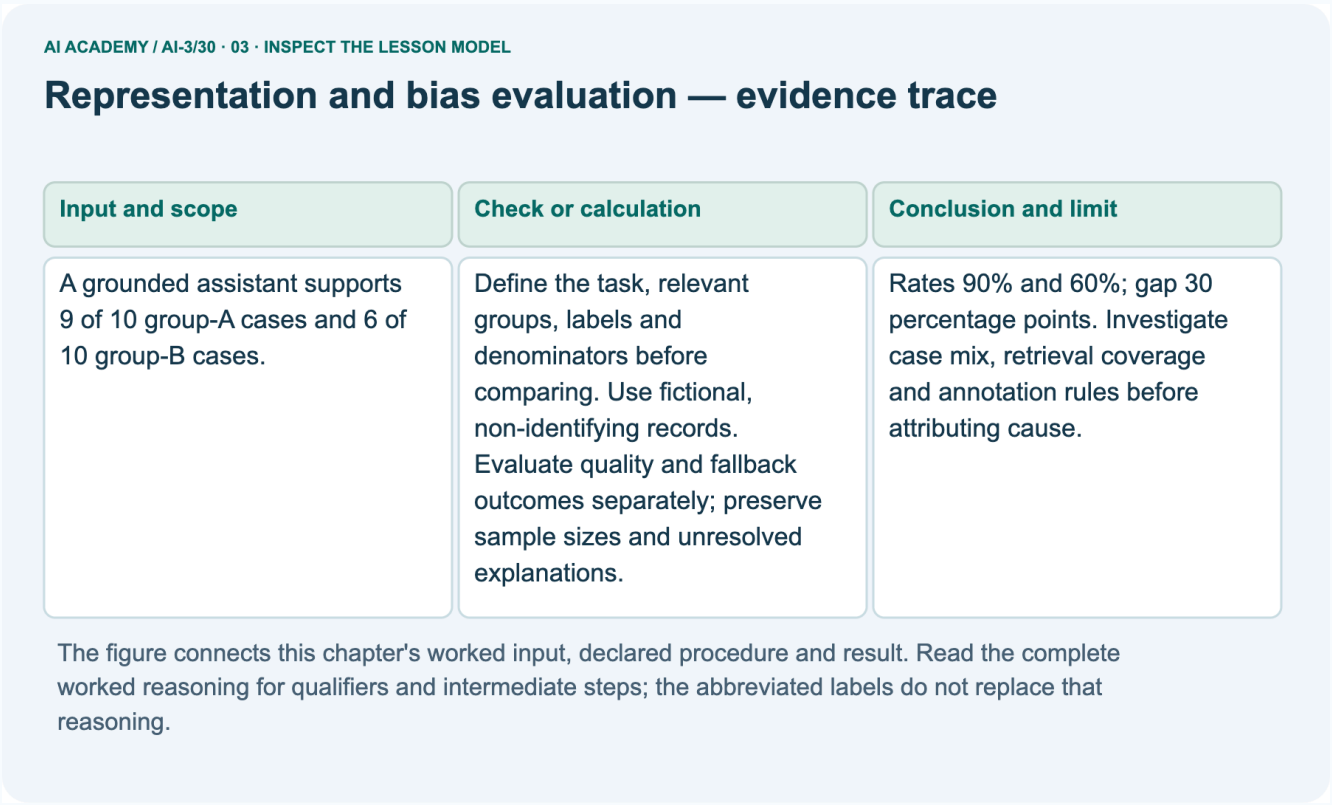
Keep denominators: Birch: 6/8.

Pine: 9/12. → Compare rates: Both support rates are 75%; Pine has more correct answers because it has more cases. → Inspect errors: Check the two Birch and three Pine failures before deciding what revision each needs.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Representation and bias evaluation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Representation and bias evaluation — **evidence** trace. A grounded assistant supports 9 of 10 group-A cases and 6 of 10 group-B cases. | Define the task, relevant groups, labels and denominators before comparing.

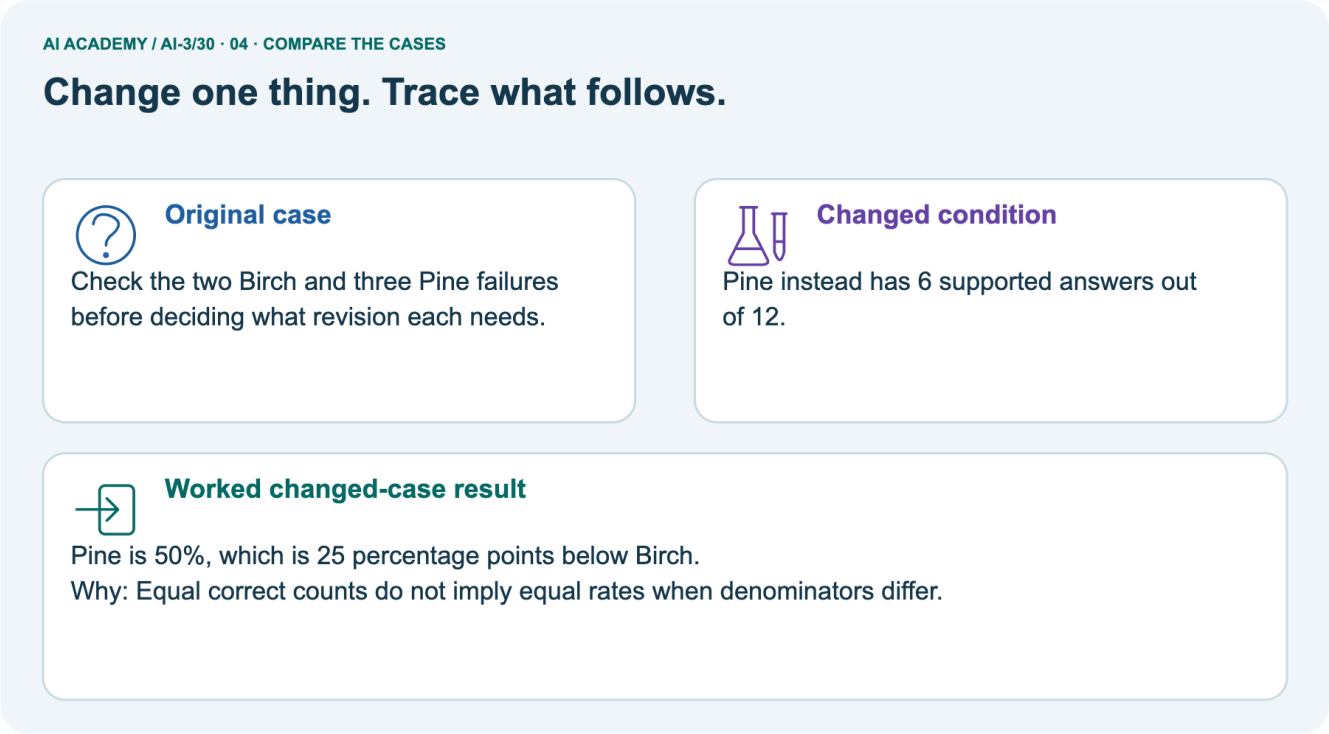
Use fictional, non-identifying records. Evaluate quality and fallback outcomes separately; preserve sample sizes and unresolved explanations. | Rates 90% and **60%**; gap 30 percentage points.

Investigate **case mix**, retrieval coverage and annotation rules before attributing cause.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

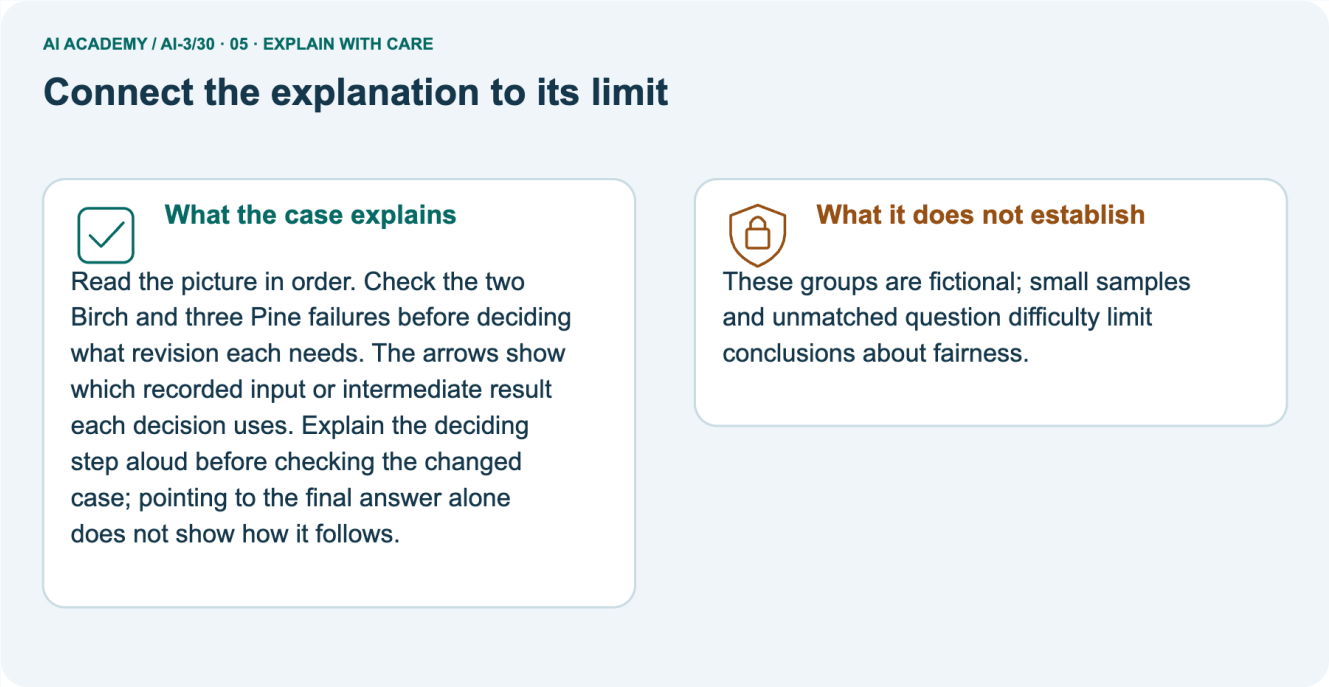
Original case: Check the two Birch and three Pine failures before deciding what revision each needs. Changed condition: Pine instead has 6 supported answers out of 12.

Expected result: Pine is 50%, which is 25 percentage points below Birch. Reason: Equal correct counts do not imply equal rates when denominators differ.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Check the two Birch and three Pine failures before deciding what revision each needs. The arrows show which recorded **input** or intermediate result each decision uses.

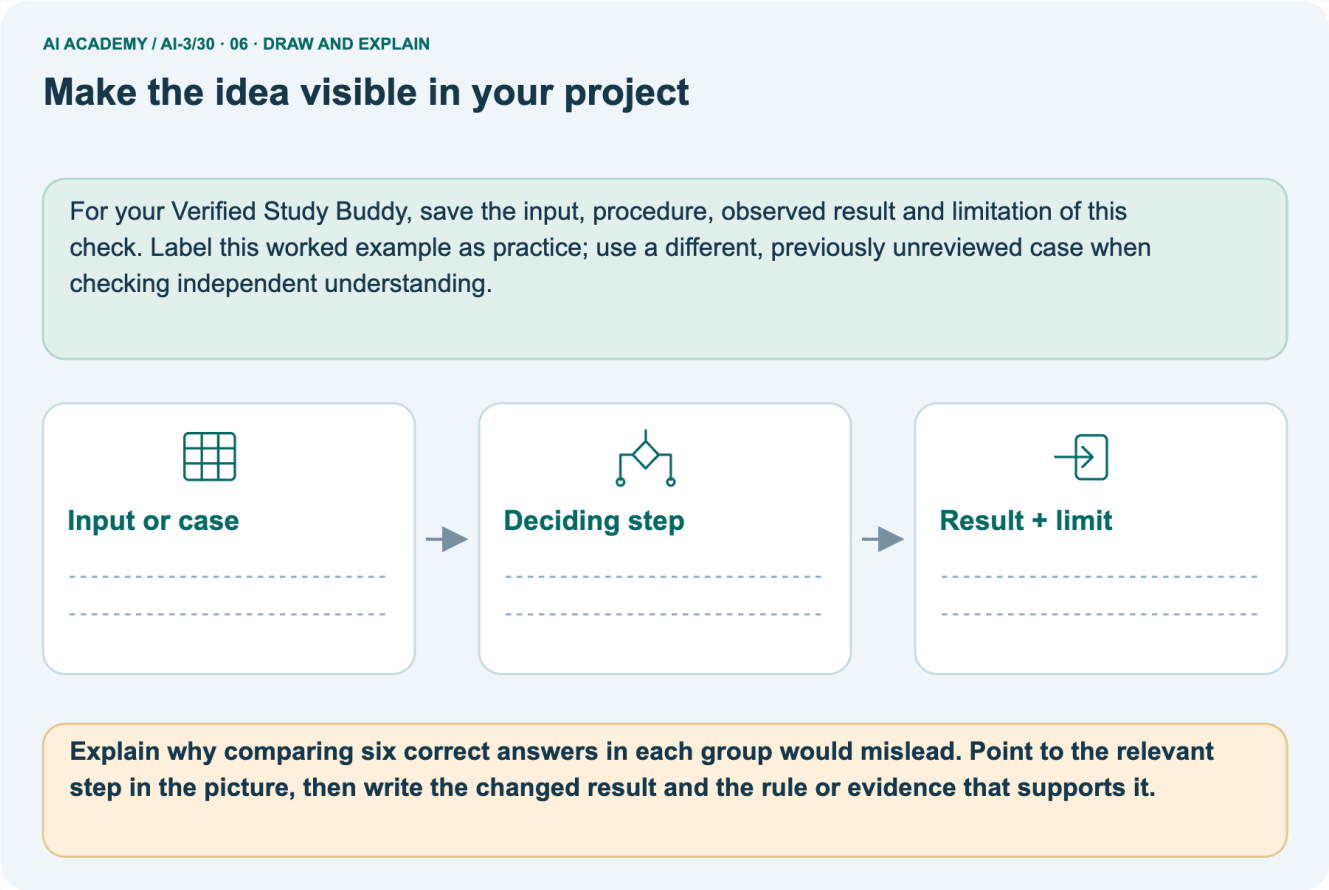
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: These groups are fictional; small samples and unmatched question difficulty limit conclusions about fairness.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Explain why comparing six correct answers in each group would mislead. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Bias and Representation: the case

02. Follow the deciding steps

03. Representation and bias evaluation — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.